

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RANCHI

Department of Computer Science & Engineering

RESEARCH INTERNSHIP REPORT

An Integrated SQL-Based Data Warehousing and Business Intelligence Framework with Machine Learning for Student Academic Risk Prediction

---

Submitted by

Yash Raj Singh

B.Tech, Computer Science & Engineering

Birla Institute of Technology, Mesra, Ranchi

Research Supervisor

Dr. Ranjan Kumar Behera

Assistant Professor, Department of CSE

IIIT Ranchi

July 2026



Abstract

The increasing volume of student data in educational institutions presents both an opportunity and a challenge for data-driven decision-making. Traditional reporting systems often fail to provide timely, actionable insights for identifying academically at-risk students. This study proposes an integrated SQL-based data warehousing and Business Intelligence (BI) framework for education analytics, augmented with a machine learning component for student academic risk prediction.

A star-schema data warehouse was designed and implemented in MySQL, comprising three dimension tables (student demographics, behavioral factors, and learning environment) and one fact table capturing academic performance metrics. An ETL pipeline was developed in Python to extract, clean, transform, and load student performance data into the warehouse. Statistical hypothesis testing — including one-way ANOVA and chi-square tests — was conducted to evaluate the significance of behavioral and demographic variables on academic outcomes.

A Random Forest classifier was trained on genuinely independent behavioral and demographic features, following a rigorous data leakage investigation that identified and excluded variables directly used in constructing the target label. The model achieved 84.84% classification accuracy, outperforming a majority-class baseline of 61.8% by 23 percentage points. Feature importance analysis revealed that study hours and age are the strongest non-linear predictors of academic risk, despite showing minimal individual correlation with performance metrics.

Results were visualized through a five-page interactive Power BI dashboard, enabling institutional stakeholders to monitor student performance, behavioral trends, and model-predicted risk categories in real time. This framework demonstrates the viability of combining data warehousing, statistical analysis, and machine learning within a unified BI pipeline for proactive academic intervention.

Keywords: Data Warehousing, Business Intelligence, Education Analytics, Random Forest, Student Risk Prediction, ETL, Power BI, Star Schema

1. Introduction

The rapid digitization of educational systems has generated unprecedented volumes of student data, encompassing academic performance, behavioral patterns, attendance records, and socioeconomic indicators. Educational institutions are increasingly recognizing the potential of this data to improve student outcomes, optimize resource allocation, and enable early identification of students at academic risk. However, the majority of institutions continue to rely on traditional, retrospective reporting systems that aggregate data only after academic performance has already declined — limiting the scope for proactive intervention.

Business Intelligence (BI) and data warehousing technologies offer a structured approach to consolidating heterogeneous educational data into a centralized, analytical repository. A well-designed data warehouse, built on a star schema, enables efficient multidimensional querying and supports the kind of real-time, interactive reporting that institutional decision-makers require. When combined with machine learning, such a framework can move beyond descriptive analytics toward predictive capabilities — identifying at-risk students before academic failure occurs, rather than after.

Existing research in education analytics has explored individual components of this pipeline in isolation — either focusing on predictive modeling using machine learning, or on dashboard-based reporting, or on data warehouse design. Few studies have proposed and validated a fully integrated, end-to-end framework that combines all three: a structured SQL data warehouse, statistical hypothesis testing for variable validation, and a leakage-aware machine learning model for risk classification, all connected to an interactive BI dashboard.

This study addresses that gap by proposing such a framework, applied to a student performance dataset comprising behavioral, demographic, and learning environment variables. The framework's machine learning component was developed with explicit attention to data leakage — a methodological concern often overlooked in education analytics research — ensuring that the predictive model relies only on genuinely independent input features rather than variables directly derived from the target label.

The primary contributions of this study are:

- Design and implementation of a star-schema SQL data warehouse for education analytics
- Statistical validation of behavioral and demographic variables using ANOVA and chi-square testing
- Development of a leakage-free Random Forest classification model achieving 84.84% accuracy, outperforming a 61.8% majority-class baseline
- A five-page interactive Power BI dashboard for real-time monitoring and predictive risk visualization

The remainder of this paper is organized as follows: Section 2 reviews related literature; Section 3 describes the methodology; Section 4 presents the results; Section 5 discusses findings and limitations; and Section 6 concludes with directions for future work.

2. Literature Review

This section reviews the body of research underpinning the proposed framework, organized into four thematic areas: education analytics, data warehousing in education, machine learning for student performance prediction, and business intelligence in education. Together, these four areas map directly onto the four components of the proposed system — data collection and early prediction, structured data storage, predictive modeling, and decision-support reporting.

2.1 Education Analytics

Lu et al. [1] applied learning analytics to the early prediction of student academic performance in a blended learning environment, demonstrating that data collected during the initial weeks of a course can be used to reliably flag students at risk of underperforming later in the term. Their work established an important precedent for this study: that prediction need not wait until final outcomes are available, and that behavioral and interim data carry genuine predictive signals.

Ghandour [2] investigated the use of learning analytics to predict student performance specifically within a Moodle Learning Management System (LMS), using LMS-logged activity data to build predictive models of academic outcomes. This study is particularly relevant to the present work in demonstrating that platform-generated behavioral data (such as activity logs, engagement indicators, and participation metrics) can serve as effective predictors independent

of raw grade data — a principle mirrored in this study's use of behavioral and environmental features rather than direct academic scores.

Alshanqiti and Namoun [3] conducted a systematic literature review of data mining and learning analytics techniques used for student performance prediction, synthesizing methods, data sources, and reported accuracy levels across a wide range of published studies. Their review highlights that model performance in this domain varies considerably based on feature selection and data quality rather than algorithm choice alone — a finding directly relevant to the data leakage investigation carried out in the present study's methodology.

Taken together, these three studies establish that (a) early, behavior-based prediction is both feasible and valuable, (b) LMS and platform-level data can substitute effectively for direct grade data, and (c) careful feature selection is a more decisive factor in prediction quality than algorithmic sophistication — a principle that directly motivated this study's rigorous separation of leakage-prone and genuinely independent features.

2.2 Data Warehousing in Education

Moscoso-Zea, Paredes-Gualtor, and Luján-Mora [4] presented a holistic view of data warehousing in education, proposing a structured, dimensional approach to consolidating institutional academic data that is otherwise scattered across disparate student information systems, learning platforms, and administrative records. Their work provides an architectural foundation for representing academic entities — students, courses, and performance events — through fact and dimension tables suited to analytical querying, directly informing the star-schema design (dim_students, dim_behavior, dim_environment, fact_performance) adopted in this study.

Barahama and Wardani [5] presented a practical data warehouse design based on Pentaho data integration to support the analysis of student learning achievement, detailing an extract-transform-load (ETL) pipeline for consolidating raw academic records into an analysis-ready schema. While this study relies on a Python-and-SQLAlchemy-based ETL pipeline rather than Pentaho, the underlying principle — that a well-designed dimensional schema is a prerequisite for reliable downstream analytics and reporting — carries over directly,

and their emphasis on structured ETL processes reinforces the truncate-and-reload approach adopted in this study's data loading procedure.

Together, these two studies confirm that a SQL-based, dimensionally modeled data warehouse is both an established and effective solution for consolidating fragmented academic data, providing the structural foundation on which the present study's statistical analysis, machine learning, and BI layers are built.

2.3 Machine Learning for Student Performance Prediction

de Andrade et al. [6] conducted a comparative study of supervised machine learning algorithms for predicting student dropout and academic success, evaluating multiple classification approaches against consistent performance metrics. Their comparative methodology — benchmarking several algorithms against a common evaluation framework — parallels this study's own comparison of a trained Random Forest classifier against a majority-class baseline to quantify genuine predictive gain.

Alashwal [7] conducted a systematic literature review of machine learning techniques applied to student performance prediction, cataloguing the range of algorithms, feature sets, and evaluation approaches used across the field, and noting that ensemble methods such as Random Forest are frequently reported among the stronger-performing approaches on structured, tabular academic datasets — a finding consistent with the algorithm choice made in this study.

Drlik [8] examined the use of different machine learning techniques specifically for predicting student dropout across university courses, comparing model performance across course types and highlighting the practical challenges of applying predictive models within real institutional workflows, including data availability and generalizability across cohorts. Although the present study targets a three-tier risk classification rather than binary dropout prediction, Drlik's findings on the practical deployment challenges of predictive models in education inform the caveats raised in this study's discussion regarding real-world applicability.

Ahmed et al. [9] conducted a meta-analysis of the literature on machine learning applications in higher education for assessing academic performance, at-risk status, and student attrition,

synthesizing findings across a large body of published studies. Their meta-analytic scope underscores the breadth and maturity of this research area while also identifying recurring gaps — including a frequent lack of transparency around feature construction and potential data leakage — that this study explicitly addresses through its dedicated leakage investigation.

Collectively, these four studies confirm the technical feasibility and established precedent for ML-based academic risk prediction, while also revealing a recurring limitation across the field: comparatively few studies rigorously interrogate their engineered features for leakage into the prediction target, a gap the present study directly addresses.

2.4 Business Intelligence in Education

Luján-Mora [10] proposed a business intelligence framework for analyzing educational data, describing how dashboards and reporting layers built atop institutional data can support administrative and academic decision-making. This work provides a conceptual basis for the multi-page Power BI dashboard developed in this study, particularly the alignment of visualized metrics with institutionally meaningful KPIs such as attendance, engagement, and performance distribution.

Scholtz, Calitz, and Haupt [11] developed a business intelligence framework for sustainability information management in higher education, focused on how BI systems can support institutional reporting on sustainability-related indicators. While their application domain differs from academic risk prediction, their framework's emphasis on aligning BI system design with clearly defined institutional reporting needs, and on the organizational factors (data governance, user readiness) that determine BI adoption success, is broadly applicable to the design considerations behind this study's dashboard.

Sorour and Atkins [12] addressed the challenge of using business intelligence dashboards to monitor quality in higher education institutions amid large and complex institutional datasets, illustrating how BI tools can distill large-scale data into actionable indicators for institutional quality assurance. Their focus on translating large, complex datasets into decision-ready visual indicators directly parallels this study's Page 5 (Predictive Risk Analysis) dashboard, which

surfaces model outputs — accuracy, feature importance, and per-class performance — in a form usable by non-technical academic stakeholders.

Together, these three studies establish that BI systems in education are most effective when purpose-built around specific institutional decision needs, reinforcing this study's approach of dedicating specific dashboard pages to distinct stakeholder concerns (executive overview, behavioral analytics, environmental factors, demographics, and predictive risk).

2.5 Synthesis and Research Gap

The reviewed literature demonstrates that (i) early, behavior-based prediction of student outcomes is a well-established and valuable practice (Lu et al., 2018; Ghandour, 2020; Alshanqiti & Namoun, 2021)[1] - [3]; (ii) SQL-based, dimensionally modeled data warehousing provides a proven architectural solution for consolidating fragmented institutional data (Moscoso-Zea et al., 2018; Barahama & Wardani, 2021)[4], [5]; (iii) a range of machine learning techniques, particularly ensemble methods, can predict academic risk and dropout with reasonable accuracy (de Andrade et al., 2024; Alashwal, 2021; Drlik, 2021; Ahmed et al., 2022)[6] - [9]; and (iv) business intelligence tools are essential for translating institutional data and predictive output into decision-ready insight (Luján-Mora, 2020; Scholtz et al., 2018; Sorour & Atkins, 2024)[10] - [12].

However, three gaps recur across this literature. First, data warehousing and BI studies in education rarely incorporate a predictive machine learning component, and ML studies rarely address the upstream data infrastructure problem, leaving these three layers largely disconnected in practice. Second, few of the reviewed ML studies rigorously examine their engineered or derived features for data leakage — the phenomenon, addressed directly in this study's methodology, where features used to construct a target label are inadvertently reused as predictors, producing artificially inflated accuracy. Third, BI-in-education studies tend to focus on descriptive institutional reporting (attendance, sustainability, quality metrics) rather than surfacing predictive model output for academic stakeholders. The present study addresses all three gaps by proposing an integrated pipeline that combines a SQL-based data warehouse, a leakage-audited machine learning model, and a BI dashboard purpose-built to surface predictive risk analysis alongside descriptive institutional metrics.

3. Methodology

This section describes the end-to-end methodology adopted in this study, structured around the actual pipeline implemented: dataset description, data preprocessing, data warehouse design, statistical analysis, a dedicated data leakage investigation, machine learning model development, and business intelligence dashboard design. This ordering reflects the genuine sequence of the research process and is intended to make the pipeline fully reproducible.

3.1 Dataset Description

This study uses the "Student Performance and Learning Behavior Dataset" sourced from Kaggle, a synthetic dataset comprising 14,003 records prior to cleaning. Each record represents an individual student and includes demographic attributes (Age, Gender), behavioral attributes (StudyHours, AssignmentCompletion, Discussions, Motivation, StressLevel), environmental attributes (Internet Access, OnlineCourses, Extracurricular, EduTech), and academic outcome attributes (Attendance, ExamScore, FinalGrade, LearningStyle). As the dataset is synthetic, it does not correspond to a real institution; it was selected for its structural realism and completeness, which allow the full data warehousing, statistical, and machine learning pipeline to be demonstrated end-to-end. This synthetic nature is treated as a limitation and is revisited in the Discussion section.

3.2 Data Preprocessing

The raw dataset was inspected for structural and categorical consistency, following which 1,534 duplicate records (10.96% of the raw data) were identified and removed using pandas' drop_duplicates(), reducing the working dataset to 12,469 records. Several columns, originally integer-encoded in the raw source, were decoded into human-readable categorical labels (for example, Gender: 0/1 → Male/Female; StressLevel and Motivation: 0/1/2 → Low/Medium/High; LearningStyle: 0/1/2/3 → Visual/Auditory/Reading-Writing/Kinesthetic) to support interpretable downstream analysis and dashboard design.

Two composite features were engineered to summarize academic and behavioral standing:

- **PerformanceIndex** = ExamScore × 0.5 + Attendance × 0.3 + AssignmentCompletion × 0.2
- **EngagementScore** = Discussions × 0.4 + OnlineCourses × 0.3 + Extracurricular × 0.3

A three-tier **RiskCategory** label was then derived from PerformanceIndex thresholds: High Risk (PerformanceIndex < 60), Medium Risk (60 ≤ PerformanceIndex < 80), and Low Risk (PerformanceIndex ≥ 80). This derivation is methodologically significant, as RiskCategory is not an independently observed outcome but a constructed label — a point that directly motivates the data leakage investigation described in Section 3.4.

The deduplication step itself proved analytically important: correlation analysis conducted prior to deduplication showed behavioral variables carrying near-zero correlation with performance, while the same analysis after deduplication revealed moderate, meaningful correlations for Attendance (0.332) and AssignmentCompletion (0.318). This indicates that duplicate records were diluting genuine behavioral signals in the raw data, underscoring deduplication as a substantive analytical step rather than a routine formality.

3.3 Data Warehouse Design

The cleaned dataset was loaded into a MySQL relational database structured as a star schema, consisting of three dimension tables and one fact table:

- **dim_students** (student_id PK, age, gender, learning_style)
- **dim_behavior** (behavior_id PK, study_hours, assignment_completion, discussions, motivation, stress_level)
- **dim_environment** (environment_id PK, internet_access, online_courses, extracurricular, edutech)
- **fact_performance** (performance_id PK, student_id FK, behavior_id FK, environment_id FK, attendance, exam_score, final_grade, performance_index, engagement_score, risk_category)

Data was loaded using Python's SQLAlchemy library. To prevent fact-table duplication on repeated pipeline runs — a subtle but consequential risk when using if_exists='append' — a

truncate-before-reload step was implemented, ensuring each run clears existing table contents before reinsertion while preserving the schema's primary key and foreign key constraints. A known limitation of the current implementation is that student_id, behavior_id, and environment_id in the fact table are assigned as simple sequential ranges rather than through explicit join keys; this works correctly only because all dimension tables are populated from the same unfiltered, unsorted source DataFrame in identical row order, and is documented in the codebase as a constraint on any future reordering of the source data.

Four SQL files support the warehouse: create_database.sql and create_tables.sql for schema creation, analytics_queries.sql containing eight analytical queries (covering risk distribution, learning style and stress-level performance breakdowns, engagement by risk category, attendance categorization, motivation analysis, and EduTech impact), and data_maintenance.sql for table truncation and row-count verification.

3.4 Statistical Analysis

Prior to model development, three statistical tests were conducted to assess the individual significance of key behavioral and demographic factors on academic performance:

- **Correlation analysis** of all numeric features against PerformanceIndex, identifying ExamScore ($r = 0.891$), Attendance ($r = 0.332$), and AssignmentCompletion ($r = 0.318$) as the strongest linear correlates, with all other features showing negligible correlation ($r < 0.03$).
- **One-way ANOVA** testing the effect of StressLevel, Motivation, and LearningStyle on PerformanceIndex. StressLevel showed a statistically significant effect ($F = 10.87$, $p < 0.001$), while Motivation ($F = 0.945$, $p = 0.389$) and LearningStyle ($F = 0.965$, $p = 0.408$) did not reach significance.
- **Chi-square test** of independence between Gender and RiskCategory, showing no statistically significant association ($\chi^2 = 4.441$, $p = 0.109$, $df = 2$).

These results directly shaped the machine learning feature set: the near-zero individual significance of Motivation and LearningStyle motivated a multivariate modeling approach

capable of capturing interaction effects that univariate tests cannot detect, rather than relying on statistically "significant" features alone.

3.5 Data Leakage Investigation

Because RiskCategory is derived directly from PerformanceIndex, and PerformanceIndex is itself a weighted function of ExamScore, Attendance, and AssignmentCompletion, an explicit leakage audit was conducted before model training. Three forms of leakage were identified:

1. **Direct (hard) leakage**: PerformanceIndex showed completely non-overlapping value ranges across RiskCategory classes (High Risk: 48–60; Medium Risk: 60–79.9; Low Risk: 80–98.8), confirming that RiskCategory could be reconstructed deterministically from PerformanceIndex alone.
2. **Soft leakage via dominant component**: ExamScore, carrying a 0.5 weight in the PerformanceIndex formula and a 0.891 correlation with it, was identified as a dominant leakage source.
3. **Soft leakage via secondary components**: Attendance and AssignmentCompletion, the remaining two components of PerformanceIndex, showed clear mean gradients across risk classes (Attendance: 70.9 / 79.4 / 84.2 for High / Medium / Low Risk; AssignmentCompletion: 64.3 / 73.2 / 79.7 respectively), confirming their leakage risk despite lower individual weights.

By contrast, EngagementScore — constructed independently from Discussions, OnlineCourses, and Extracurricular, none of which contribute to PerformanceIndex — showed overlapping, near-identical distributions across all three risk classes (range approximately 0–12.4, means clustering around 6.0–6.2), confirming it as a safe, leakage-free feature.

Based on this audit, PerformanceIndex, ExamScore, Attendance, and AssignmentCompletion were excluded from the model's feature set. This step is treated as a central methodological contribution of this study, distinguishing it from prior work in this domain that frequently omits explicit leakage verification of engineered or derived features.

3.6 Machine Learning Model Development

A Random Forest classifier (scikit-learn, n_estimators=100, random_state=42) was trained to predict RiskCategory using the eleven remaining, leakage-free features: Age, Gender, StudyHours, Discussions, Motivation, StressLevel, Internet, OnlineCourses, Extracurricular, EduTech, and EngagementScore. Categorical variables were properly encoded before training. The dataset was split 80/20 into training (9,975 records) and test (2,494 records) sets. Model performance was benchmarked against a majority-class DummyClassifier baseline to establish a genuine measure of predictive gain, and evaluated using accuracy, per-class precision/recall/F1-score, a confusion matrix, and Random Forest feature importance scores. Model predictions and feature importance values were exported to CSV for integration into the Power BI dashboard.

3.7 Business Intelligence Dashboard Design

A five-page Power BI dashboard was developed to translate the statistical, warehousing, and modeling outputs into an interactive decision-support tool. Page 1 (Executive Overview) presents institution-level KPIs and risk distribution; Page 2 (Behavioral Analytics) and Page 3 (Learning Environment Analysis) break down performance by behavioral and environmental factors respectively; Page 4 (Demographics & Risk Analysis) examines risk distribution across age and gender; and Page 5 (Predictive Risk Analysis) surfaces the machine learning model's output directly, including model accuracy, per-class accuracy, feature importance, and a confusion matrix visual. Each page includes cross-filtering by Risk Category and relevant behavioral or environmental dimensions, allowing stakeholders to interactively explore both descriptive institutional patterns and predictive model behavior within a single tool.

4. Results

This section presents the findings of the study in a purely descriptive form, organized to mirror the methodology: data preprocessing outcomes, statistical test results, data leakage audit findings, machine learning model performance, and dashboard outputs. Interpretation of these results is deferred to the Discussion section.

4.1 Data Preprocessing Outcomes

The raw dataset contained 14,003 records. Deduplication removed 1,534 duplicate records (10.96% of the raw data), yielding a cleaned working dataset of 12,469 records. No missing values were present in the cleaned dataset following this step.

Stage	Record Count
Raw dataset	14,003
Duplicates removed	1,534
Cleaned dataset	12,469

Table 4.1 — Dataset size before and after cleaning

4.2 Statistical Analysis Results

4.2.1 Correlation Analysis

Feature	Correlation with Performance Index
Exam Score	0.891
Attendance	0.332
Assignment Completion	0.318
All other numeric features	< 0.03

Table 4.2 — Correlation of numeric features with PerformanceIndex (post-deduplication)

4.2.2 ANOVA Results

Factor	F-Statistic	P-value	Result (α = 0.05)
Stress Level	10.87	0.0000192	Significant
Motivation	0.945	0.389	Not Significant
Learning Style	0.965	0.408	Not Significant

Table 4.3 — One-way ANOVA results for behavioral/demographic factors on PerformanceIndex

4.2.3 Chi-Square Test

Statistic	Value
χ^2 (Chi - square)	4.441
df	2
P-value	0.109
Result (α = 0.05)	Not Significant

Table 4.4 — Chi-square test of independence: Gender vs. RiskCategory

4.3 Data Leakage Audit Findings

Risk Category	Performance Index Range
High Risk	48 - 60
Medium Risk	60 - 79.9
Low Risk	80 - 98.8

Table 4.5 — PerformanceIndex value ranges by RiskCategory

Feature	High Risk	Medium Risk	Low Risk
Attendance	70.9	79.4	84.2
Assignment Completion	64.3	73.2	79.7
Engagement Score	~6.0	~6.1	~6.2

Table 4.6 — Mean values of leakage-audited features by RiskCategory

EngagementScore range was approximately 0 – 12.4 across all three RiskCategory classes, with overlapping distributions and near-identical class means, in contrast to the non-overlapping ranges observed for PerformanceIndex, Attendance, and AssignmentCompletion.

Final feature set retained for modeling (11 features): Age, Gender, StudyHours, Discussions, Motivation, StressLevel, Internet, OnlineCourses, Extracurricular, EduTech, EngagementScore.

Features excluded due to leakage: PerformanceIndex, ExamScore, Attendance, AssignmentCompletion.

4.4 Machine Learning Model Performance

Split	Record Count
Training set	9,975
Test set	2,494

Table 4.7 — Train/test split

Model	Accuracy
Majority-class baseline (Dummy Classifier)	61.8%
Random Forest Classifier	84.84%
Improvement	+23.04 percentage points

Table 4.8 — Model accuracy vs. baseline

Class	Precision	Recall	F1-Score	Support
High Risk	0.88	0.53	0.66	184
Medium Risk	0.84	0.95	0.89	1,542
Low Risk	0.87	0.72	0.79	768
Weighted Avg	**0.85**	**0.85**	**0.84**	**2,494**

Table 4.9 — Classification report (Random Forest, test set)

Actual\Predicted	High Risk	Medium Risk	Low Risk	Total
High Risk	97	75	12	184
Medium Risk	9	1,465	68	1,542
Low Risk	4	210	554	768
Total	**110**	**1,750**	**634**	**2,494**

Table 4.10 — Confusion matrix (Actual vs. Predicted)

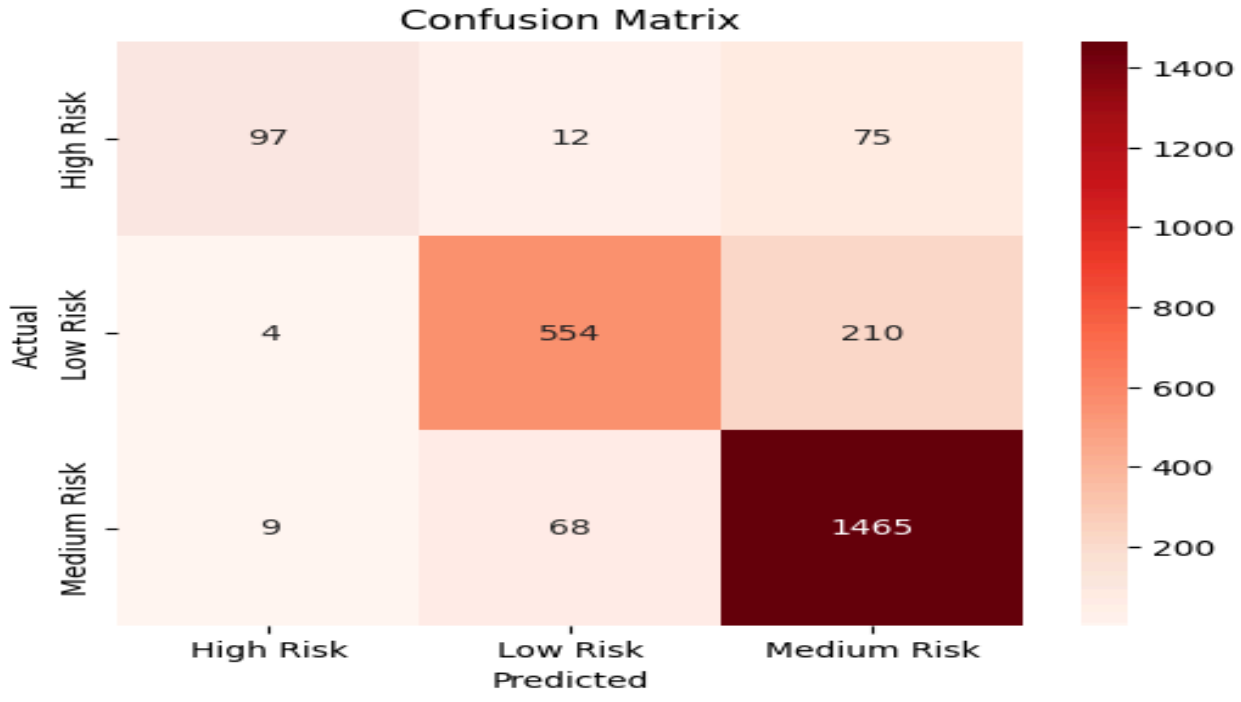


Figure 4.1: *Confusion matrix of Random Forest predictions on the test set*

Rank	Feature	Importance
1	Study Hours	0.2665
2	Age	0.1985
3	Engagement Score	0.1342
4	Online Courses	0.1229
5	Stress Level	0.0713
6	Motivation	0.0535
7	Gender	0.0482
8	Edu Tech	0.0376
9	Extracurricular	0.0335
10	Discussions	0.0170
11	Internet	0.0169

Table 4.11 — Random Forest feature importance (sorted, descending)

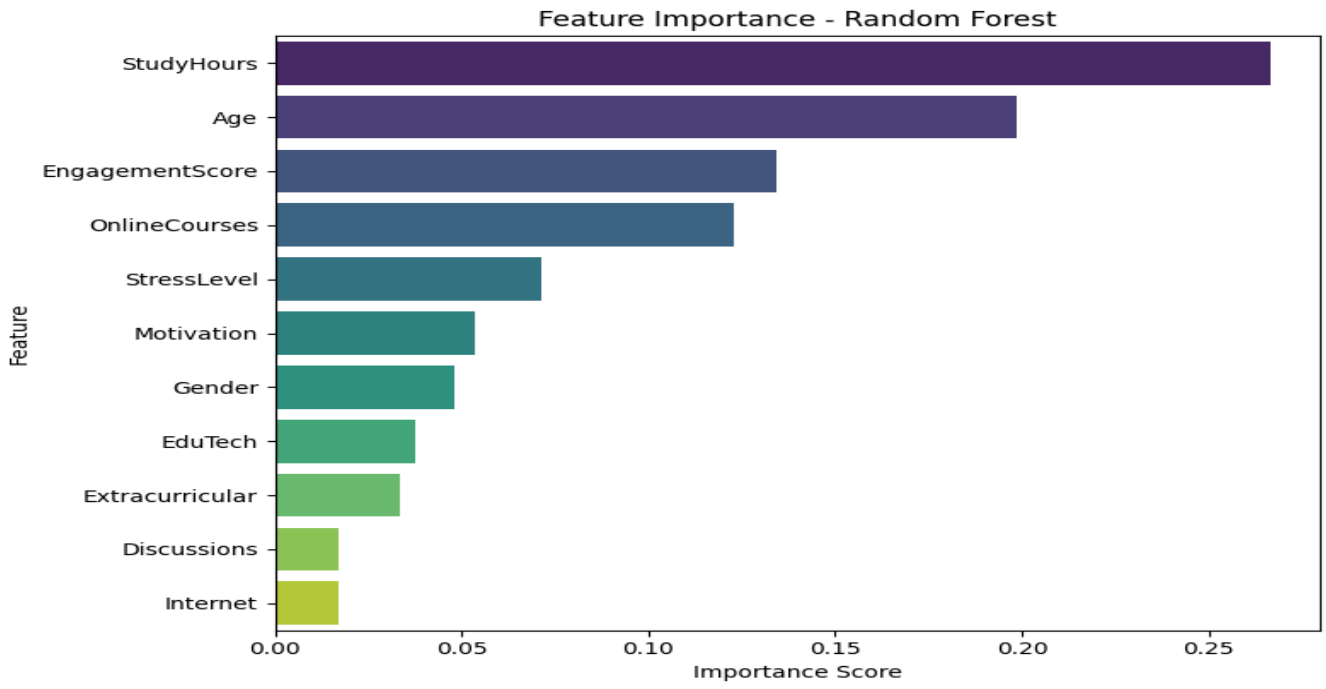


Figure 4.2: *Feature importance scores from the Random Forest classifier*

4.5 Business Intelligence Dashboard Output

The five-page Power BI dashboard produced the following summary indicators, drawn directly from the cleaned dataset and model outputs:

KPI	Value	Dashboard Page
Post-Deduplication Records	12,469 (shown as 12K)	Page 1
Average Performance Score	74.13	Page 1
Average Engagement Score	6.17	Page 1
Average Attendance (%)	80.24	Page 1
Average Study Hours	20.03	Page 2
Assignment Completion (%)	74.52	Page 2
Average Discussions	0.61	Page 2
Average Exam Score	70.31	Page 3
Internet Access Rate	91.71	Page 3
Average Online Courses Taken	9.87	Page 3
High EduTech Adoption Rate (%)	70.72	Page 3
Average Age	23.53	Page 4
High Risk Student (%)	7.91%	Page 4
Female Student (%)	53.86%	Page 4
Model Accuracy	84.84%	Page 5
Total Predictions (test set)	2,494	Page 5
Correct Predictions	2,116	Page 5

Table 4.12 — Selected dashboard KPIs

Figures 4.3–4.7 present the five dashboard pages, each translating the statistical, warehousing, and modeling outputs described above into an interactive view for institutional stakeholders.

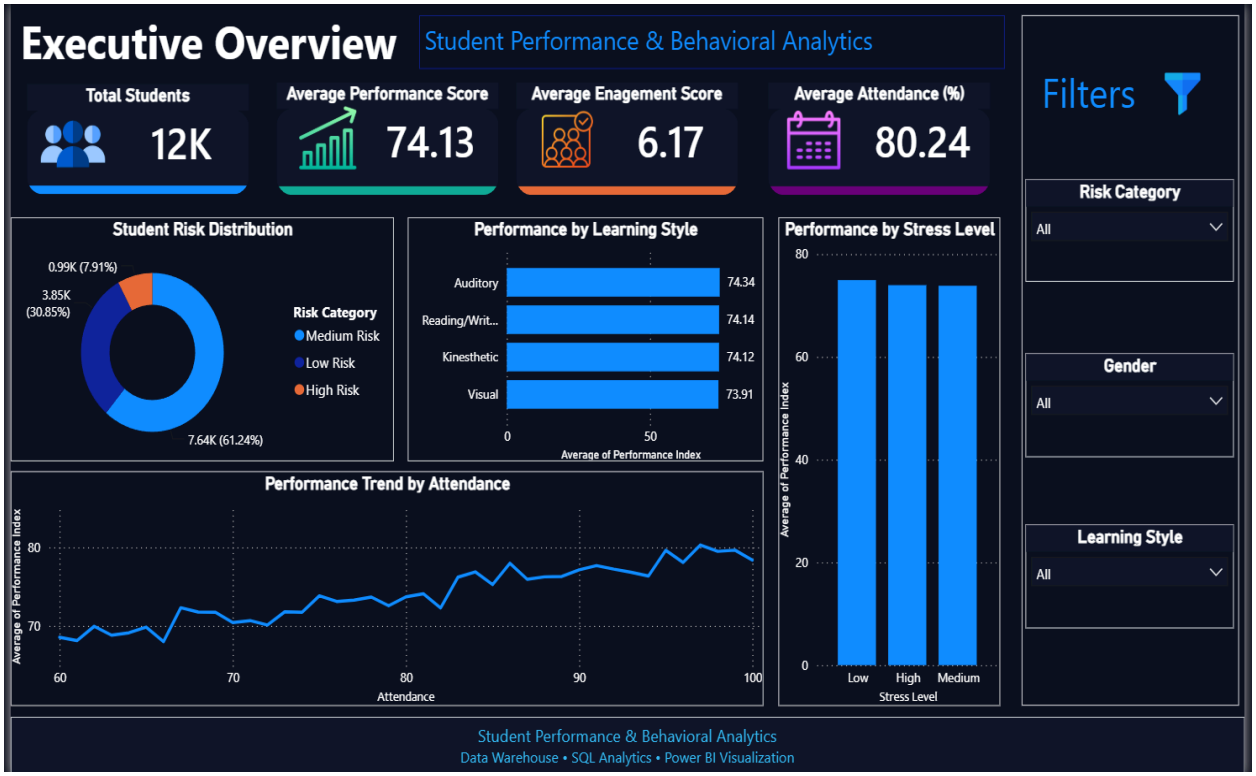


Figure 4.3: *Page 1: Executive Overview*



Figure 4.4: *Page 2: Behavioral Analytics*

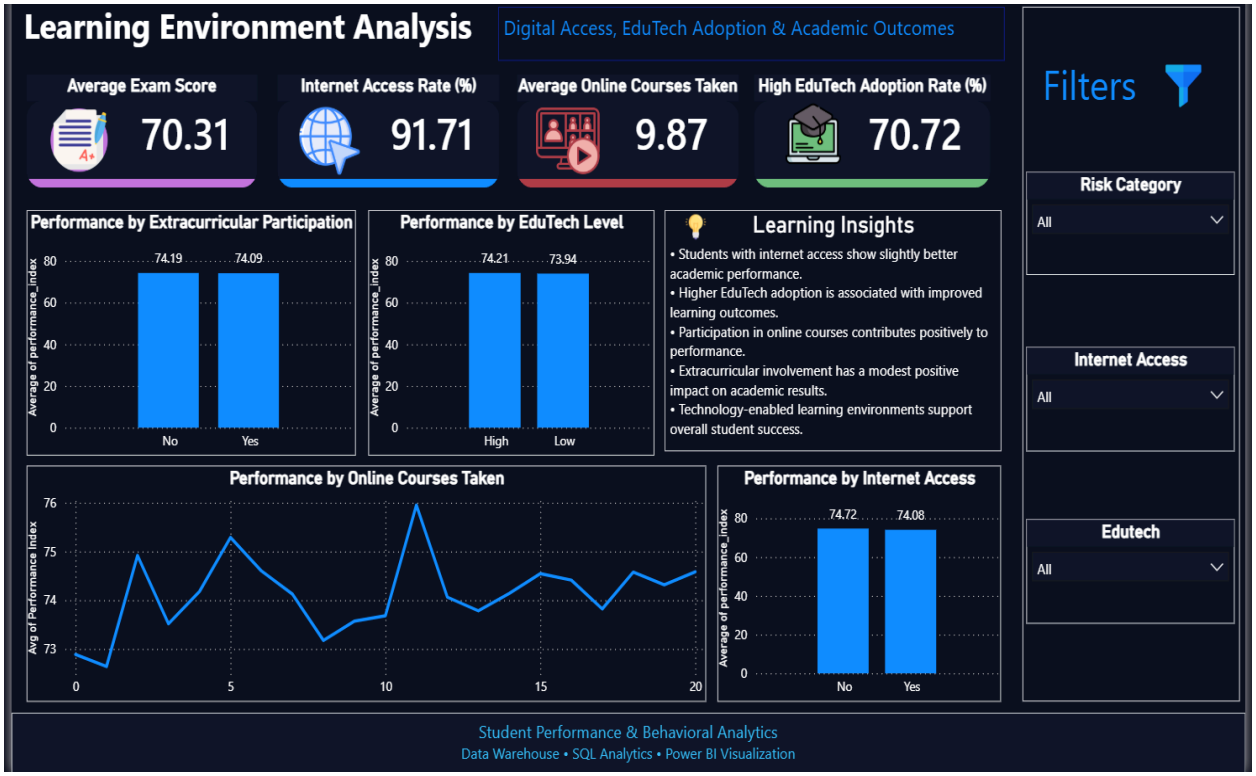


Figure 4.5: *Page 3: Learning Environment Analysis*

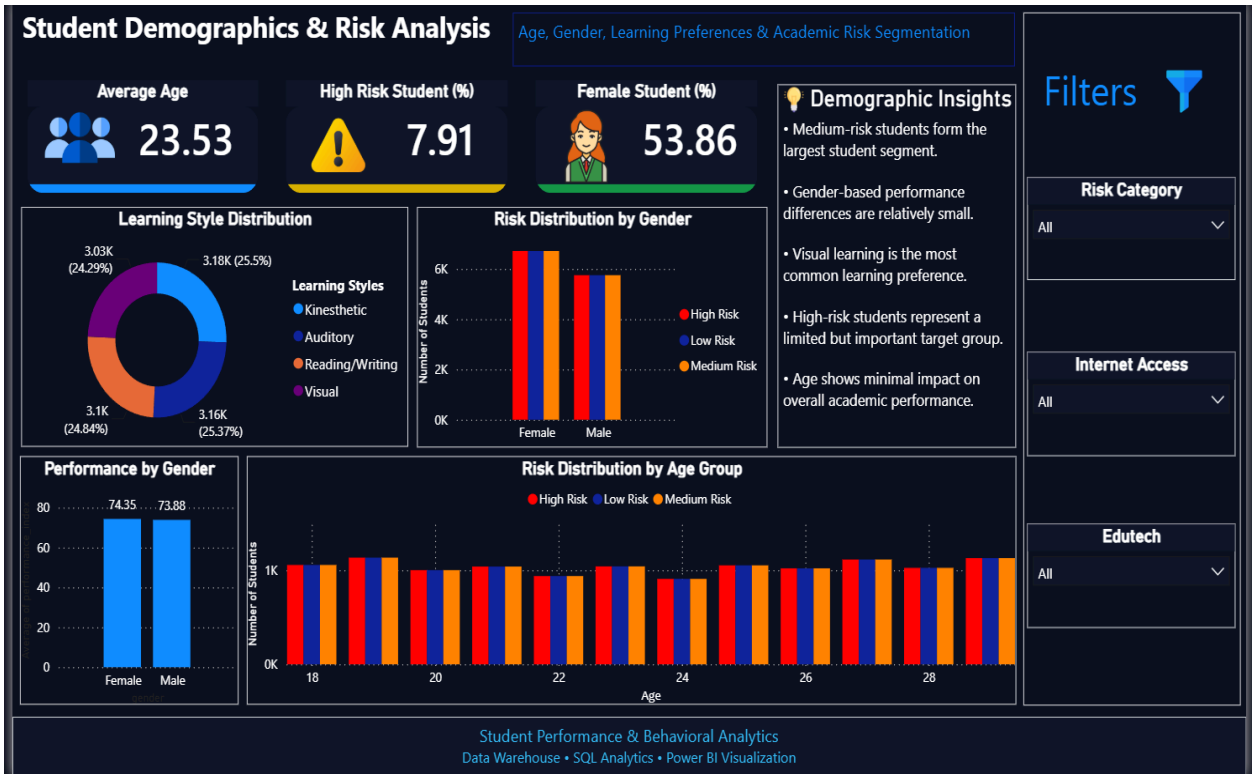


Figure 4.6: *Page 4: Student Demographics & Risk Analysis*



Figure 4.7: *Page 5: Predictive Risk Analysis*

5. Discussion

This section interprets the findings presented in Section 4, connecting them back to the literature reviewed in Section 2 and to the objectives outlined in the Introduction. The discussion is organized around five themes: the significance of deduplication, the divergence between correlation and feature importance, the implications of the data leakage findings, the practical limitations of the predictive model, and the broader contribution of the integrated framework.

5.1 The Analytical Importance of Deduplication

The finding that behavioral variables showed near-zero correlation with performance before deduplication, but moderate and interpretable correlation afterward (Attendance: 0.332, AssignmentCompletion: 0.318), indicates that duplicate records were not a neutral data quality issue but an active source of analytical distortion. Duplicated rows effectively diluted genuine

behavioral signals by inflating the apparent variance and noise in the dataset. This finding reinforces a point made only implicitly in the data warehousing literature reviewed in Section 2.2 (Moscoso-Zea et al., 2018; Barahama & Wardani, 2021)[4], [5]: that the structural quality of a data pipeline — not just its architecture — directly determines the validity of any analysis built on top of it. A well-designed star schema loaded with duplicated or poorly cleaned data would have yielded the same underestimated correlations observed here, underscoring that data warehouse design and data quality assurance must be treated as inseparable concerns.

5.2 Correlation versus Predictive Importance: Evidence of Non-Linear Interaction Effects

One of the more analytically interesting findings of this study is the divergence between individual statistical significance and machine learning feature importance. StudyHours and Age showed negligible individual correlation with PerformanceIndex (well under the 0.03 threshold observed for most non-leaking features), and Motivation and LearningStyle failed to reach significance in ANOVA testing. Yet in the Random Forest model, StudyHours (importance = 0.2665) and Age (importance = 0.1985) emerged as the two strongest predictors of RiskCategory, together accounting for nearly 47% of total feature importance.

This divergence is consistent with the observation made by Alshanqiti and Namoun [3] that model performance in student prediction tasks depends heavily on the modeling approach's ability to capture patterns beyond what univariate statistical tests can detect. A plausible explanation is that StudyHours and Age contribute to RiskCategory not through a direct linear relationship with performance, but through interaction effects — for instance, the effect of study hours on outcomes may differ systematically by age group, or by combination with stress and engagement levels, in ways that a single Pearson correlation coefficient cannot capture but a tree-based ensemble model can. This finding has a direct methodological implication: relying solely on correlation or ANOVA results to perform feature selection prior to modeling — a common simplification in some of the reviewed literature — would have led to the exclusion of the two most predictive variables in this dataset, illustrating a genuine risk in premature statistical feature elimination.

5.3 Implications of the Data Leakage Findings

The data leakage investigation described in Section 3.5 and evidenced in Section 4.3 represents a central methodological contribution of this study. The identification of PerformanceIndex, ExamScore, Attendance, and AssignmentCompletion as leakage sources — through non-overlapping value ranges and clear mean gradients across RiskCategory classes — demonstrates that a model trained on the full, unaudited feature set would have reported inflated and practically meaningless accuracy, since it would essentially be reconstructing a label from the same components used to define it rather than genuinely predicting an independent outcome.

This finding is significant in light of Ahmed et al.'s [9] meta-analytic observation that transparency around feature construction is frequently lacking in the broader machine learning-for-education literature. The present study's explicit leakage audit — distinguishing "hard" leakage (deterministic reconstruction) from "soft" leakage (strong but non-deterministic correlation via shared formula components) — offers a template that could be applied more broadly in similar studies using engineered or derived target labels. The confirmation that EngagementScore, constructed independently of the PerformanceIndex formula, showed genuinely overlapping distributions across risk classes further validates the audit methodology: the absence of a gradient in a construction-independent feature serves as an internal control, supporting confidence that the leakage identified in the other features was real and not an artifact of the audit method itself.

5.4 Practical Limitations of the Predictive Model

While the Random Forest model's 84.84% accuracy represents a substantial improvement over the 61.8% majority-class baseline, per-class performance reveals an important practical limitation: High Risk recall was only 53%, meaning nearly half of genuinely high-risk students in the test set were misclassified — predominantly into the Medium Risk category, as shown in the confusion matrix (75 of 184 High Risk cases predicted as Medium Risk). In a real early-warning intervention context, this is a meaningful shortcoming, since High Risk students

are precisely the group for whom correct identification carries the greatest practical consequence, and a 53% detection rate would miss a substantial proportion of students most in need of support.

This limitation is likely attributable to class imbalance: High Risk students constituted only 184 of 2,494 test cases (7.4%), compared to 1,542 Medium Risk cases (61.8%), giving the model comparatively little exposure to High Risk examples during training. This observation is consistent with the findings in [8] on the practical deployment challenges of predictive models in education, where class imbalance and cohort-specific data availability were identified as recurring obstacles to reliable dropout and risk prediction in applied settings. Future iterations of this framework could address this through class-weighting, oversampling techniques such as SMOTE, or threshold adjustment specifically calibrated to prioritize High Risk recall over overall accuracy, given the asymmetric cost of false negatives in this class.

A second limitation concerns the synthetic nature of the underlying dataset. Because RiskCategory is a derived analytical label rather than an independently observed outcome (such as actual dropout or failure), the reported accuracy should be interpreted as a demonstration of the framework's technical capability — spanning data warehousing, statistical validation, leakage-audited modeling, and BI integration — rather than as evidence of real-world predictive validity. Deployment on genuine institutional data, with an independently defined and observed risk or dropout outcome, would be required before the framework's predictive claims could be considered practically validated.

5.5 Contribution of the Integrated Framework

Positioned against the literature reviewed in Section 2, this study's central contribution is architectural rather than purely algorithmic: it demonstrates a complete, internally consistent pipeline connecting data warehousing, statistical validation, leakage-audited machine learning, and BI-based decision support — the very integration identified as absent in Section 2.5's synthesis of the four reviewed research areas. Where Moscoso-Zea et al. [4] and Barahama and Wardani [5] address warehousing without predictive modeling, and Luján-Mora [10], Scholtz et al. [11] , and Sorour and Atkins [12] address BI reporting largely independent of a predictive layer, this study's Page 5 dashboard directly surfaces model accuracy, feature importance, and

per-class performance alongside the descriptive institutional metrics on Pages 1–4, within a single coherent tool grounded in a shared, leakage-audited data pipeline.

The divergence between correlation and feature importance (Section 5.2) and the explicit leakage audit (Section 5.3) further suggest that methodological rigor of this kind — rarely made explicit in the reviewed education analytics literature — is necessary not merely as good practice, but as a precondition for trustworthy predictive claims in this domain. Any future extension of this framework to genuine institutional data should preserve both practices: multivariate, non-linear modeling rather than reliance on univariate feature screening, and explicit auditing of any engineered or derived target labels for leakage prior to model training.

6. Conclusion

This study set out to design and implement an integrated SQL-based data warehousing and business intelligence framework, augmented with machine learning, for predicting student academic risk. The work combined four components that are rarely addressed together in the existing literature: a dimensionally modeled SQL data warehouse, rigorous statistical validation of behavioral and demographic factors, a leakage-audited machine learning classifier, and a multi-page BI dashboard surfacing both descriptive institutional metrics and predictive model output.

The results demonstrate that a Random Forest classifier trained on eleven genuinely independent behavioral, demographic, and environmental features achieved 84.84% accuracy in predicting student risk category, a substantial improvement over the 61.8% majority-class baseline. Statistical testing confirmed stress level as the only individually significant behavioral predictor of performance ($p < 0.001$), while motivation and learning style showed no significant individual effect — yet the machine learning model revealed that StudyHours and Age, despite negligible individual correlation with performance, were the two strongest predictors overall, highlighting the presence of non-linear interaction effects that univariate statistical methods alone cannot detect. A dedicated data leakage audit, identified as a central methodological contribution of this study, uncovered that four of the original fifteen features were directly or indirectly derived from

the target label itself, and their exclusion was essential to obtaining a genuinely valid model rather than an artificially inflated one.

The study's primary contribution is architectural: it demonstrates that data warehousing, statistical analysis, machine learning, and business intelligence reporting can be integrated into a single, internally consistent pipeline, addressing a fragmentation observed across the reviewed literature, where these four areas are typically studied and published in isolation from one another. The explicit auditing of engineered features for data leakage, in particular, offers a methodological practice that could be adopted more broadly in similar education analytics research.

The study also has clear limitations, most notably its reliance on a synthetic dataset and a derived, rather than independently observed, risk label, alongside a class-imbalance-driven weakness in detecting High Risk students (53% recall). These limitations mean the framework's predictive results should be read as a demonstration of technical capability rather than as validated real-world predictive performance. Future work should focus on three directions: validating the framework on genuine institutional data with an independently observed outcome variable (such as actual course failure or dropout); addressing class imbalance through techniques such as SMOTE or class-weighted training to improve High Risk detection specifically; and extending the BI layer to support real-time or near-real-time data refresh, enabling the framework to function as an operational early-warning tool rather than a static analytical demonstration.

Overall, this study contributes a reproducible, methodologically rigorous template for combining SQL-based data warehousing, statistically grounded feature analysis, leakage-audited machine learning, and business intelligence reporting into a single framework for education analytics — one that, with validation on real institutional data, could support genuine early identification of academically at-risk students and inform timely, data-driven intervention.

References

[1] Lu, O. H. T., Huang, A. Y. Q., Huang, J. C. H., Lin, A. J. Q., Ogata, H., & Yang, S. J. H. (2018). Applying learning analytics for the early prediction of students' academic performance in blended learning. Educational Technology & Society, 21(2), 220–232. https://www.jstor.org/stable/26388400

[2] Ghandour, A. (2020). Using learning analytics to predict students' performance in Moodle LMS. International Journal of Emerging Technologies in Learning (iJET), 15(20). https://doi.org/10.3991/ijet.v15i20.15915

[3] Alshanqiti, A., & Namoun, A. (2021). Predicting student performance using data mining and learning analytics techniques: A systematic literature review. Applied Sciences, 11(1), 237. https://doi.org/10.3390/app11010237

[4] Moscoso-Zea, O., Paredes-Gualtor, J., & Luján-Mora, S. (2018). A holistic view of data warehousing in education. IEEE Access, 6. https://doi.org/10.1109/ACCESS.2018.2866796

[5] Barahama, A. D., & Wardani, R. (2021). Data analysis and data warehouse design based on Pentaho data integration to support student learning achievement. IOP Conference Series: Materials Science and Engineering, 1098(5), 052089. https://doi.org/10.1088/1757-899X/1098/5/052089

[6] de Andrade, C. R. V., et al. (2024). Supervised machine learning algorithms for predicting student dropout and academic success: A comparative study. Discover Artificial Intelligence, 4(1), 2. https://doi.org/10.1007/s44163-023-00079-z

[7] Alashwal, H. (2021). A systematic literature review of student performance prediction using machine learning techniques. Education Sciences, 11(9), 552. https://doi.org/10.3390/educsci11090552

[8] Drlik, M. (2021). Towards predicting student's dropout in university courses using different machine learning techniques. Applied Sciences, 11(7), 3130. https://doi.org/10.3390/app11073130

[9] Ahmed, K., et al. (2022). Application of machine learning in higher education to assess student academic performance, at-risk, and attrition: A meta-analysis of literature. Education and Information Technologies, 27(3), 3743–3775. https://doi.org/10.1007/s10639-021-10741-7

[10] Luján-Mora, S. (2020). A business intelligence framework for analyzing educational data. Sustainability, 12(14), 5745. https://doi.org/10.3390/su12145745

[11] Scholtz, B., Calitz, A., & Haupt, R. (2018). A business intelligence framework for sustainability information management in higher education. International Journal of Sustainability in Higher Education, 19(2), 266–290. https://doi.org/10.1108/IJSHE-06-2016-0118

[12] Sorour, A., & Atkins, A. S. (2024). Big data challenge for monitoring quality in higher education institutions using business intelligence dashboards. Journal of Electronic Science and Technology, 22(1), 100233. https://doi.org/10.1016/j.jnlest.2024.100233